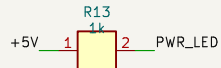
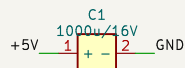
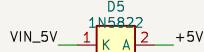
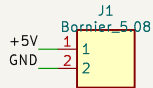
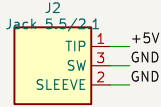
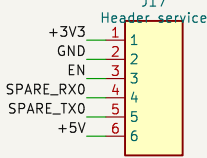


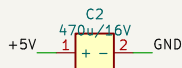
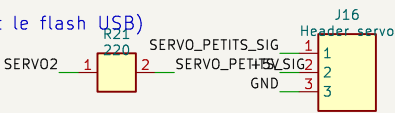
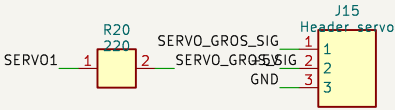
ALIMENTATION 5V (3A recommandé) – jack OU bornier.
D5 protège le rail si l'USB du DevKit est branché en même temps.



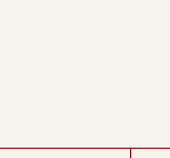
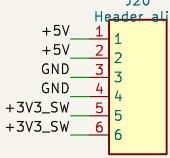
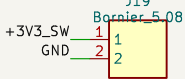
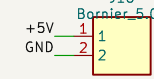
TOUS les GPIO libres sur header (Dupont),
y compris RX0/TX0 (les laisser libres pendant le flash USB)
et GPIO36/39 (entrees seules).



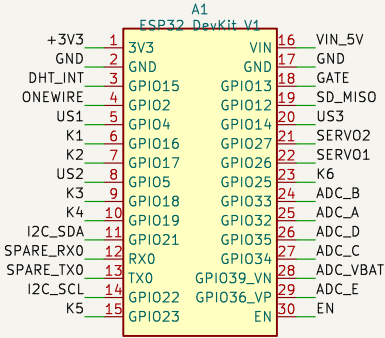
SERVOS NOURRISSEURS 5V
GPIO12 (strapping MTDI) : pull-down R22, signal série R20.



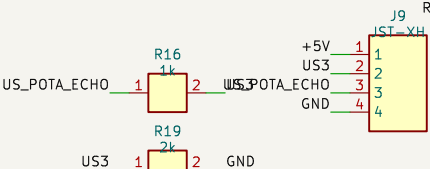
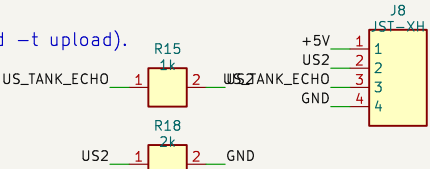
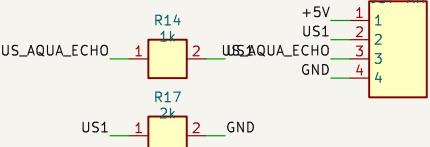
Distribution 5V / 3V3–capteurs / GND :
borniers a yis + rail header.



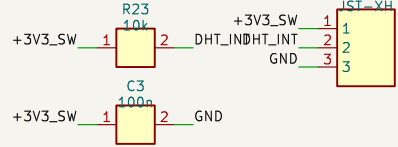
ESP32 DevKit V1 (30 broches, socketé).
Flash/monitor via l'USB du module (pio run -e wroom-prod -t upload).



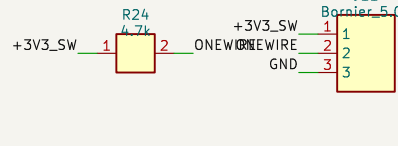
3x HC-SR04 (5V), mode mono–broche TRIG=ECHO (sensor_ultrasonic.cpp).
Echo 5V ramené à 3V3 par pont 1k/2k sur chaque canal.



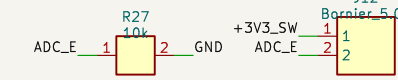
AIR : DHT11 (option DHT22, –DUSE_DHT22) – pull-up 10k.



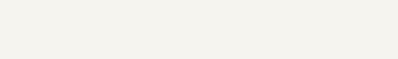
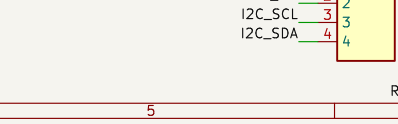
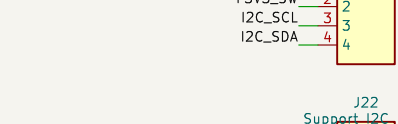
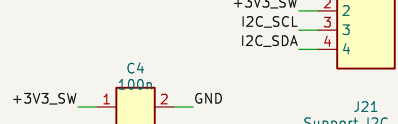
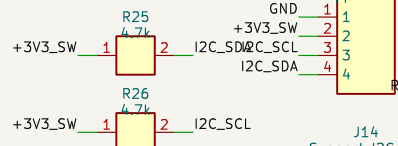
EAU : DS18B20 (1–Wire, pull-up 4.7k).



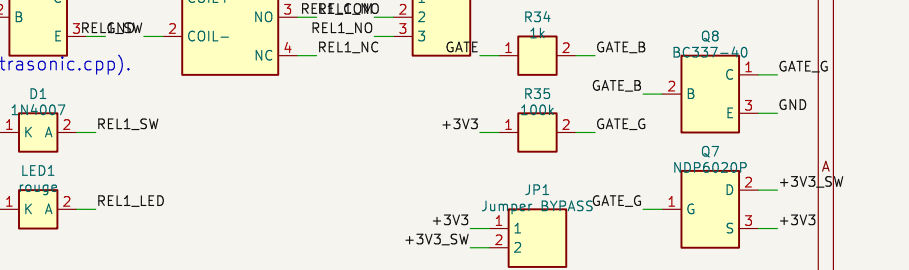
LUMIERE : LDR déportée + 10k (GPIO34 = entrée seule).



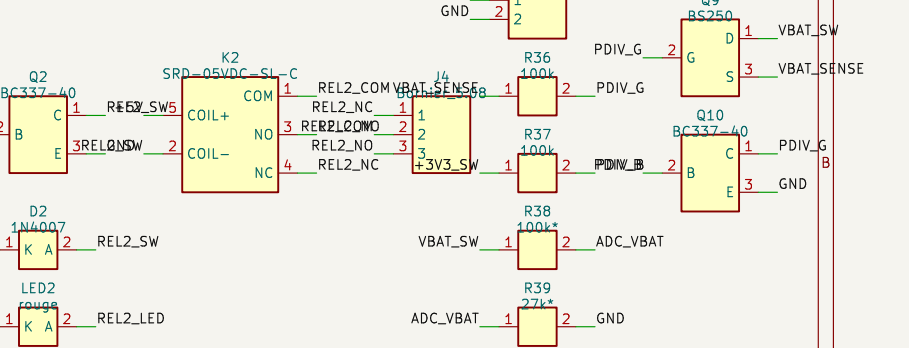
I2C : OLED SSD1306 0x3C + header extension (DS3231 si besoin).



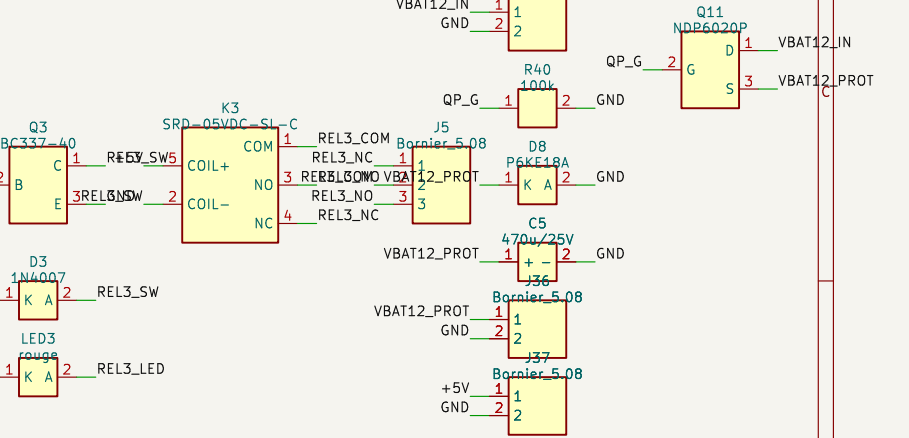
A POWER-TEG EN AMONT (fusible/disjoncteur) : charges inductives : snubber cote charge.
POWER–GATE +3V3_SW (GPIO13, topologie n3pp–JP1 FERME par défaut (rail permanent, ffp5cs) :
profil batterie –> tout le rail capteurs est coupé



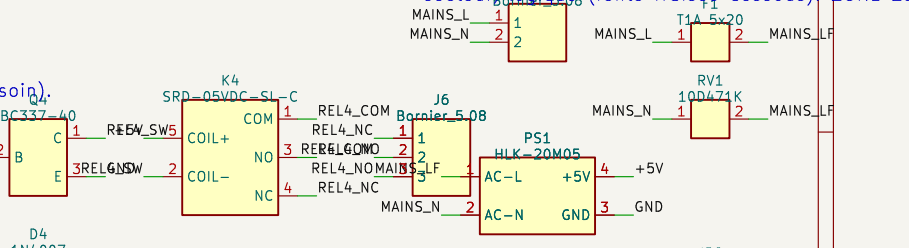
PONT DIVISEUR VBAT COMMUTE : actif seulement
R38/R39 SELON PROFIL : 1S Li-ion = 100k/100k



PROFIL BUS 12V SOLAIRE : fusible lame 7,5–10A
anti-inversion P–MOSFET, TVS 18V, réservoir ; bu
(module MP1584/XL4015 faible Iq sur entretoise



PROFIL SECTEUR : Hi–Link HLK–20M05
+ varistance 300VAC. Le corps du module enjam
secteur logique (fente fraisee dessous) ZONE 23



microSD : module SPI 3V3 (B2) : JP2/3/4 : se
CS/CLK/MISO en 5V3 (1–2 default) ou WROO
MISO = 1 net paragee exteactua1–GPIO12 / A2–IO1

